

Slope and Rate of Change: Find the Error

Directions

- Slope is $m = \frac{\text{change in } y}{\text{change in } x}$. Take both differences *in the same order*, starting from the same point.
- In each “find the error” problem, work the slope out yourself first, then compare it with what the student wrote.
- When you name an error, say which quantity ended up in the wrong place — “upside down” and “sign flipped” are different mistakes.

1. Find the slope of the line through $A(2, 1)$ and $B(6, 9)$.

Answer: _____

2. Jonah found the slope of the line through $P(-3, 4)$ and $Q(1, -4)$. He wrote $m = \frac{1 - (-3)}{-4 - 4} = -\frac{1}{2}$.

- (a) Find the correct slope.

Answer: _____

- (b) Name the error Jonah made.

Answer: _____

3. Mara found the slope of the line through $R(5, 2)$ and $S(1, 10)$. She wrote $m = \frac{10 - 2}{5 - 1} = 2$.

(a) Find the correct slope.

Answer: _____

(b) Name the error Mara made.

Answer: _____

4. A tank is draining. The table shows the gallons of water left after t hours. Ravi says the rate of change is $-\frac{6}{18}$ gallons per hour.

Hours	0	2	4	6
Gallons left	40	34	28	22

(a) How many gallons does the tank lose between hour 0 and hour 6?

Answer: _____

(b) Find the rate of change in gallons per hour.

Answer: _____

(c) Name the error in Ravi's calculation.

Answer: _____

5. A printer prints 300 pages in 12 minutes. Sam wants the slope of the graph of pages against minutes and writes $m = \frac{12}{300} = 0.04$.

(a) Find the correct slope, in pages per minute.

Answer: _____

(b) Name the error Sam made.

Answer: _____

6. The points $(1, 9)$ and $(7, 3)$ are on a skateboard ramp. Aisha wrote $m = \frac{9 - 3}{7 - 1} = 1$.

(a) Find the correct slope.

Answer: _____

(b) Name the error Aisha made.

Answer: _____

7. A booth sells tickets. The table shows the total cost in dollars for different numbers of tickets. Leo says the slope is $\frac{6}{36}$ dollars per ticket.

Tickets	2	4	6	8
Total cost	15	27	39	51

(a) How much more do 8 tickets cost than 2 tickets?

Answer: _____

(b) Find the slope, in dollars per ticket.

Answer: _____

(c) Name the error in Leo's calculation.

Answer: _____

8. Dana and Kofi both computed the slope of the line through $(-2, 5)$ and $(4, -7)$. Dana got -2 ; Kofi got 2 .

(a) Compute the slope yourself.

Answer: _____

(b) Explain which student is right and what the other one must have done.

Answer: _____